

Topological Analysis of RNA Splicing in Endometriosis

Integrating Genetic Susceptibility, Environmental Response, and Hodge Theory

BINF 2111 Project Proposal

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1 Project Overview

Endometriosis is a complex disease that likely cannot be explained by one gene or one environmental factor. Genetic susceptibility, environmental exposures, changes in gene regulation, and differences in gene expression may all contribute to the disease.

A common bioinformatics approach would be to identify genes associated with endometriosis, compare them with genes affected by environmental exposures, and study where those lists overlap.

We want to take this idea one step further.

Instead of only asking:

Which genes are associated with endometriosis?

or:

Which genes are expressed differently in endometriosis?

we want to ask:

Are the same genes being processed differently at the RNA-splicing level in endometriosis?

Our project will combine two ideas:

1. Identify genes that may be especially relevant to endometriosis because of genetic susceptibility and/or environmental response.
2. Analyze the RNA-splicing structure of those genes using graph theory and Hodge decomposition.

The final computational product would be a program written primarily in Rust that compares splice-junction networks between endometriosis and control samples and identifies genes whose splicing structure changes the most.

2 The Main Research Question

Our main research question is:

Do genes associated with endometriosis show differences in the topology of RNA splicing between endometriosis and normal endometrial tissue?

We are especially interested in genes that fall into one or both of the following categories:

1. genes associated with inherited susceptibility to endometriosis,
2. genes whose expression or regulation has been associated with environmental exposures.

The goal is not simply to find whether these genes are expressed more or less.

Instead, we want to determine whether the **way their RNA transcripts are assembled** differs between disease and control samples.

3 Why Combine Genetics and Environmental Response?

Endometriosis is unlikely to have a single simple cause.

One possible model is:

Genetic Susceptibility + Environmental Influence + Regulatory Changes $\rightarrow$ Disease Phenotype

Some individuals may carry genetic variants that increase susceptibility.

Environmental factors may also influence gene expression or regulation.

These two effects may interact.

We can define two initial gene sets:

$$G_{\text{risk}} = \{\text{genes associated with genetic risk}\}$$

and

$$G_{\text{env}} = \{\text{genes associated with environmental response}\}.$$

We can then examine:

$$G_{\text{overlap}} = G_{\text{risk}} \cap G_{\text{env}}.$$

Genes in this overlap may be especially interesting because they connect inherited susceptibility with environmentally responsive biology.

However, we do not have to limit the project only to the overlap.

We can compare three groups:

1. genetic-risk genes,
2. environmentally responsive genes,
3. genes appearing in both groups.

4 Important Clarification About Environmental Exposure

If our RNA-sequencing dataset does not directly contain information about patient exposure to a particular pesticide, chemical, or environmental factor, we cannot claim that an environmental exposure caused a splicing change.

Instead, environmental-response data would initially be used to **prioritize genes**.

For example:

Exposure-associated gene

combined with

Endometriosis-associated gene

would identify a gene that may be especially interesting to investigate.

We would then ask whether that gene also shows altered RNA-splicing structure in endometriosis.

A stronger causal analysis could be performed later if a dataset containing both exposure information and RNA-sequencing data is available.

5 Why RNA Splicing?

Gene expression usually measures how much RNA is produced from a gene.

However, the same gene can produce multiple RNA transcripts.

This happens through a process called **alternative splicing**.

Suppose a gene contains four exons:

$$E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_4.$$

One transcript might contain all four exons.

Another transcript might skip exon 2:

$$E_1 \rightarrow E_3 \rightarrow E_4.$$

Another may use a different splice site.

In some cases, a transcript may contain a back-splice junction such as:

$$E_3 \rightarrow E_1.$$

This can create a cyclic transcript structure.

Therefore, two samples can have similar expression levels for a gene while still producing different transcript structures.

That difference may be biologically important.

6 The Computational Idea

For every gene we analyze, we will construct a **splice graph**.

The graph is represented as:

$$G = (V, E),$$

where:

- each vertex represents an exon or splice site,
- each directed edge represents an observed splice junction,
- each edge has a value representing how strongly that junction is used.

For example:

$$E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_4$$

would be represented by edges:

$$(E_1, E_2), \quad (E_2, E_3), \quad (E_3, E_4).$$

Alternative splicing may add:

$$(E_1, E_3).$$

Back-splicing may add:

$$(E_3, E_1).$$

The complete splicing pattern therefore becomes a mathematical network.

7 From RNA-Seq Reads to Edge Values

RNA-sequencing can provide counts describing how many reads support each splice junction.

For example:

From	To	Supporting Reads
E_1	E_2	183
E_2	E_3	141
E_3	E_4	129
E_1	E_3	32
E_3	E_1	47

We should be careful about calling raw read counts “flow.”

RNA-seq counts are measurements of junction usage rather than literal physical flow.

Therefore, the program will first convert junction counts into normalized edge values.

For example, if the total junction support for a gene is

$$T = \sum_{e \in E} c_e,$$

then the normalized junction usage for edge e could be

$$w_e = \frac{c_e}{T},$$

where c_e is the read count supporting that junction.

The resulting vector

$$F = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix}$$

will be treated as the edge signal used in the mathematical analysis.

More advanced normalization can be added later if necessary.

8 Why Use Hodge Theory?

A normal graph algorithm can tell us whether a cycle exists.

However, that is not the only question we care about.

We want to determine how the **observed junction usage is distributed across different mathematical structures in the graph.**

This is where Hodge theory becomes useful.

Hodge decomposition allows an edge signal to be separated into mathematically different components.

The basic idea is:

$$\text{observed edge signal} = \text{directional component} + \text{cyclic component}.$$

If we later construct a higher-order simplicial complex, the cyclic part can be separated further into local curl and global harmonic components.

9 The Incidence Matrix

Suppose our graph contains n vertices and m edges.

We construct an incidence matrix

$$B_1 \in \mathbb{R}^{n \times m}.$$

Each column represents one directed edge.

For an edge

$$E_i \rightarrow E_j,$$

we place:

$$-1$$

at the source vertex,

$$+1$$

at the destination vertex,

and zero everywhere else.

For example, the three-edge cycle

$$E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_1$$

has the incidence matrix

$$B_1 = \begin{bmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}.$$

This matrix converts the biological graph into an object that can be analyzed using linear algebra.

10 The Gradient Component

The first part of the decomposition asks whether the edge signal can be explained by a potential assigned to the vertices.

Let

$$\phi$$

represent values assigned to the exons or splice sites.

The corresponding gradient edge signal is

$$B_1^T \phi.$$

We find the potential that best explains our observed junction signal by solving:

$$\phi^* = \arg \min_{\phi} \|F - B_1^T \phi\|_2^2.$$

The gradient component is then:

$$F_{\text{grad}} = B_1^T \phi^*.$$

This represents the portion of the splice-junction pattern that can be explained using a directional potential over the graph.

11 The Important Correction: Residual Does Not Automatically Mean Cyclic

After calculating the gradient component, we obtain:

$$F_{\text{residual}} = F - F_{\text{grad}}.$$

However,

$$F_{\text{residual}} \neq \text{automatically cyclic flow.}$$

A non-gradient residual only tells us that the observed edge pattern cannot be completely explained by a gradient.

Therefore, we should not simply calculate the residual and immediately call it “cyclic.”

Instead, we must analyze which mathematical subspace the residual belongs to.

This is an important part of the project.

12 Cycle Space

For a graph, cyclic edge signals lie in the kernel of the incidence matrix:

$$\ker(B_1).$$

An edge signal h belongs to the cycle space when

$$B_1 h = 0.$$

In simple terms, this represents edge structure that circulates through the graph rather than behaving like a simple directional gradient.

Therefore, for the first version of the project, we can use the graph-level Hodge decomposition:

$$F = F_{\text{grad}} + F_{\text{cycle}}.$$

The program can calculate the projection of the observed edge signal onto:

$$\text{im}(B_1^T)$$

and

$$\ker(B_1).$$

This gives us a mathematically justified measurement of directional versus cyclic structure.

13 Optional Full Hodge Decomposition

A more complete Hodge decomposition requires higher-order structures.

If we construct a simplicial complex, we introduce a second boundary matrix:

$$B_2.$$

The matrices have different meanings:

- B_1 describes how edges connect to vertices.
- B_2 describes how two-dimensional faces connect to edges.

The full edge-space decomposition becomes:

$$F = B_1^T \phi + B_2 \psi + h.$$

The three components are:

$$B_1^T \phi$$

for gradient flow,

$$B_2 \psi$$

for local curl-like structure,

and

$$h$$

for harmonic structure.

The harmonic component satisfies:

$$B_1 h = 0$$

and

$$B_2^T h = 0.$$

The corresponding first Hodge Laplacian is:

$$L_1 = B_1^T B_1 + B_2 B_2^T.$$

The harmonic subspace is related to:

$$\ker(L_1).$$

For the first version of the project, we do **not** need to force a B_2 matrix into the model unless we can define biologically meaningful two-dimensional structures.

Therefore:

Graph-level Hodge decomposition will be the core method. Full simplicial Hodge decomposition will be treated as an extension.

14 Why the Laplacian Matters

A Laplacian is a matrix that summarizes relationships across a graph.

The ordinary graph Laplacian can be written as:

$$L_0 = B_1 B_1^T.$$

Instead of analyzing one edge at a time, the Laplacian captures how the complete network is connected.

The Hodge Laplacian extends this idea to edge relationships.

Rather than asking:

Does this one splice junction look unusual?

we can ask:

Does the entire pattern of junction usage contain a mathematically unusual global structure?

15 Comparing Endometriosis and Control Samples

For each gene, we will construct a splice graph for endometriosis samples and control samples.

Let

$$F_g^{(E)}$$

represent the splice-junction signal for gene g in endometriosis.

Let

$$F_g^{(C)}$$

represent the splice-junction signal for the same gene in control tissue.

We perform the decomposition separately:

$$F_g^{(E)} = F_{\text{grad},g}^{(E)} + F_{\text{cycle},g}^{(E)}$$

and

$$F_g^{(C)} = F_{\text{grad},g}^{(C)} + F_{\text{cycle},g}^{(C)}.$$

16 A Simple Topological Measurement

One quantity we could calculate is the fraction of edge-signal energy lying in the cycle space.

For gene g :

$$C_g = \frac{\|F_{\text{cycle},g}\|_2^2}{\|F_g\|_2^2}.$$

This does not have to be treated as a new biological index.

It is simply a mathematical summary produced by the decomposition.

We can compare:

$$C_g^{(E)}$$

and

$$C_g^{(C)}.$$

The difference is:

$$\Delta C_g = C_g^{(E)} - C_g^{(C)}.$$

A large positive value would indicate that the gene contains more cyclic splice structure in the endometriosis samples than in the controls.

17 Example

Imagine a gene has the following structure in control tissue:

$$E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_4.$$

Its decomposition might be dominated by gradient structure.

Now imagine that the endometriosis samples contain additional strongly supported junctions:

$$E_1 \rightarrow E_3$$

and

$$E_3 \rightarrow E_1.$$

The graph now contains:

$$E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_1.$$

The cycle-space component may increase.

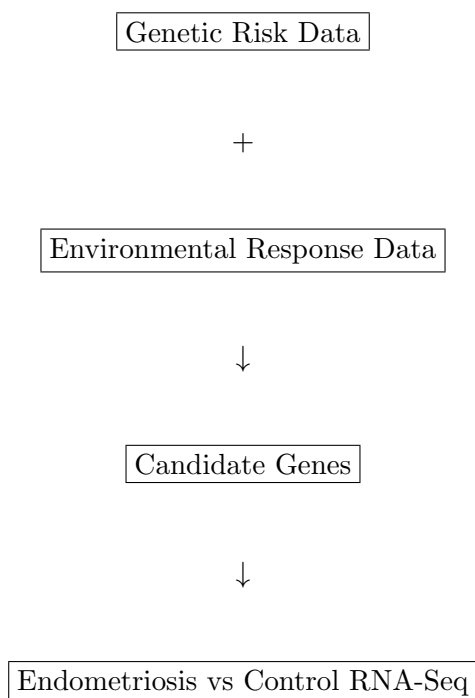
The important result would not simply be:

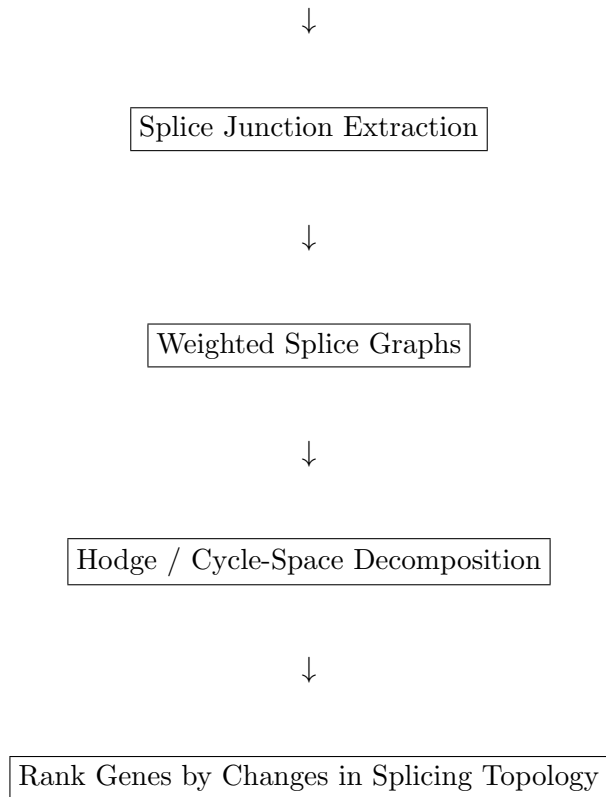
“A back-splice junction exists.”

Instead, we can quantify how much the **overall splicing architecture** changes between the two conditions.

18 Overall Computational Pipeline

The complete project can be represented as:





19 What the Rust Program Would Do

The main computational product would be a Rust command-line program.

A possible command could be:

```
cargo run -- compare control.tsv endometriosis.tsv
```

The input could contain:

```
gene,from,to,count
GENE_A,E1,E2,183
GENE_A,E2,E3,141
GENE_A,E3,E4,129
GENE_A,E3,E1,47
```

The program would:

1. read the splice-junction file,
2. group junctions by gene,

3. normalize junction usage,
4. construct directed graphs,
5. construct incidence matrices,
6. calculate gradient projections,
7. calculate cycle-space projections,
8. compare disease and control samples,
9. rank genes by topological differences,
10. identify the junctions contributing most strongly to those differences.

20 Example Program Output

The output could look like:

Comparing:

Control vs Endometriosis

Genes analyzed: 247

Gene	Control Cycle	Endo Cycle	Difference
GENE_A	0.04	0.31	+0.27
GENE_B	0.08	0.09	+0.01
GENE_C	0.02	0.24	+0.22
GENE_D	0.11	0.06	-0.05

Largest topological change:

GENE_A

Major junctions contributing to cycle structure:

E1 -> E2

E2 -> E3

E3 -> E1

These numbers are only examples of how the output could be displayed.

They are not experimental results.

21 Why Rust?

Rust is useful for this project because the program may eventually need to process many genes and many splice junctions.

Important computational tasks include:

- parsing large biological files,
- graph construction,
- sparse matrix operations,
- graph traversal,
- projection onto linear subspaces,
- parallel analysis of many genes.

Rust provides:

- fast execution,
- memory safety,
- efficient data structures,
- support for parallel computation,
- strong type checking.

The first version can operate on a small dataset, while the same architecture can later scale to hundreds or thousands of genes.

22 Possible Rust Project Structure

A possible structure is:

```
src/  
  main.rs  
  parser.rs  
  genes.rs  
  graph.rs  
  matrix.rs  
  hodge.rs  
  compare.rs  
  ranking.rs  
  output.rs
```

```
parser.rs
```

Reads input files.

`genes.rs`

Groups splice junctions by gene and tracks gene categories.

`graph.rs`

Constructs weighted directed splice graphs.

`matrix.rs`

Constructs incidence matrices and other required matrices.

`hodge.rs`

Performs the mathematical decomposition.

`compare.rs`

Compares endometriosis and control samples.

`ranking.rs`

Ranks genes according to topological changes.

`output.rs`

Produces readable results.

23 Research Aims

Aim 1: Identify biologically relevant candidate genes

Construct groups of genes associated with:

- endometriosis genetic susceptibility,
- environmental response,
- both genetic susceptibility and environmental response.

Aim 2: Build splice-junction graphs

Use RNA-sequencing data from endometriosis and control endometrial tissue to create weighted splice graphs for candidate genes.

Aim 3: Quantify splicing topology

Use graph-level Hodge decomposition to calculate directional and cyclic components of each splice graph.

Aim 4: Compare disease and control topology

Identify genes showing the largest change in splice topology between endometriosis and control samples.

24 Hypothesis

Our initial hypothesis is:

Genes involved in endometriosis will show differences not only in expression level but also in RNA-splicing topology, and these differences may be especially pronounced among genes associated with both genetic susceptibility and environmental response.

The purpose of the project is to test this hypothesis.

We do not assume that it will be correct.

25 Testing the Mathematical Method First

Before using real endometriosis data, we should validate the computational method using synthetic graphs.

Test 1: Linear Splicing

$$E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_4.$$

This should contain little or no cycle-space structure.

Test 2: Exon Skipping

$$E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_4$$

with

$$E_1 \rightarrow E_3.$$

This introduces alternative splicing without automatically creating a directed cycle.

Test 3: Back-Splicing

$$E_1 \rightarrow E_2 \rightarrow E_3$$

with

$$E_3 \rightarrow E_1.$$

This creates a clear cycle.

Test 4: Noise

Small random junction signals can be added to test whether weak sequencing noise falsely produces large cycle-space measurements.

Test 5: Changing Edge Weights

We can keep the same graph but gradually increase the weight of a back-splice junction.

For example:

$$w(E_3, E_1) = 0.01, 0.05, 0.10, 0.20, 0.40.$$

We can then test whether the calculated cycle component responds appropriately.

This directly demonstrates why the junction weights matter.

26 Real Data

The project can use publicly available data rather than generating new biological samples.

We would need several types of information:

1. a list of genes associated with endometriosis genetic susceptibility,
2. a list of genes associated with a selected environmental exposure or class of exposures,
3. RNA-sequencing data from endometriosis tissue,
4. RNA-sequencing data from appropriate control tissue.

We do not need to analyze every gene in the genome during development.

The large dataset gives us the ability to perform a substantial analysis, while candidate-gene filtering keeps the first implementation manageable.

27 Big Data Without a Slow Live Demonstration

The full analysis could eventually contain:

many patients $\times$ thousands of genes $\times$ many splice junctions.

However, the classroom demonstration does not need to process the complete dataset.

For example, we could demonstrate:

```
cargo run -- gene GENE_A
```

and show the complete decomposition for one gene.

The larger analysis could be run beforehand.

This allows the project to use biologically meaningful amounts of data without making the live demonstration slow.

28 Statistical Comparison

If multiple biological samples are available, we do not need to compare only one average control graph with one average disease graph.

For every individual sample s , we could calculate:

$$C_{g,s},$$

the cycle-space fraction for gene g in sample s .

We would then have distributions:

$$\{C_{g,s}^{(E)}\}$$

for endometriosis and

$$\{C_{g,s}^{(C)}\}$$

for controls.

We could compare these groups using a statistical test or permutation procedure.

A simple permutation approach would:

1. calculate the observed difference between groups,
2. randomly shuffle disease/control labels,
3. recalculate the difference,
4. repeat many times,

5. determine how unusual the real difference is.

This avoids relying completely on a particular assumed statistical distribution.

29 What Makes the Combined Project Different

A simpler version of the biological project would be:

$$\text{genetic genes} \cap \text{environment genes} = \text{interesting genes}.$$

A simpler RNA-seq project would be:

$$\text{endometriosis} \rightarrow \text{differential expression}.$$

Our project instead asks:

Which biologically relevant genes change

in the mathematical organization of their splicing?

The biological component tells us **where to look**.

The computational component tells us **what new property to measure**.

30 What Would Count as Success?

The project would be successful if we can show that:

1. genetic-risk and environmental-response information can be used to define candidate genes,
2. RNA splice-junction data can be converted into weighted graphs,
3. the graphs can be represented using incidence matrices,
4. the edge signal can be separated into gradient and cycle-space components,
5. synthetic cyclic and non-cyclic graphs produce distinguishable results,
6. the program can compare endometriosis and control samples,
7. the program can identify genes with unusually large changes in splicing topology,
8. at least one small example can run in real time.

31 Minimum Viable Project

If the full biological analysis becomes too large for the course timeline, the minimum version would still include:

1. select a small set of endometriosis-relevant genes,
2. load splice-junction data for endometriosis and controls,
3. build one graph per gene,
4. normalize edge weights,
5. construct B_1 ,
6. calculate the gradient projection,
7. calculate the cycle-space projection,
8. compare disease and control graphs,
9. rank the genes,
10. demonstrate the program in Rust.

This would still test the central research question.

32 Possible Extensions

If the first version works, several extensions are possible.

Environmental Exposure Data

If patient-level exposure data become available, we could directly compare:

low exposure

and

high exposure

rather than only using exposure-associated genes for prioritization.

Full Simplicial Hodge Decomposition

If we can define biologically meaningful higher-order splice structures, we could introduce B_2 and separate cyclic structure into curl and harmonic components.

Genome-Wide Analysis

Instead of analyzing only candidate genes, the tool could eventually process all genes and search for unexpected disease-associated changes in splice topology.

Other Diseases

The computational framework would not be limited to endometriosis.

The same approach could be applied to:

- cancer,
- neurological disease,
- autoimmune disease,
- developmental disorders,
- other diseases involving abnormal RNA processing.

33 Limitations

Several limitations are important.

First, a graph cycle does not automatically prove the existence of a circular RNA molecule.

Sequencing errors, alignment errors, low coverage, and ambiguous reads can affect splice-junction measurements.

Second, a difference in splicing topology does not automatically mean that the difference causes endometriosis.

Third, genes associated with environmental response do not prove that a specific environmental exposure caused the observed disease pattern.

Fourth, the mathematical usefulness of Hodge decomposition for this problem must be demonstrated rather than assumed.

The method could potentially show no meaningful advantage over simpler splice-junction measurements.

That is still a valid research result.

34 Why This Is Computational Biology

The biology includes:

- endometriosis,

- genetic susceptibility,
- environmental response,
- gene expression,
- alternative splicing,
- RNA sequencing.

The mathematics includes:

- linear algebra,
- graph theory,
- vector spaces,
- matrix projections,
- null spaces,
- graph Laplacians,
- Hodge theory,
- topology.

The computation includes:

- large-file parsing,
- graph construction,
- sparse matrices,
- numerical linear algebra,
- graph algorithms,
- statistical comparison,
- ranking algorithms,
- Rust programming.

The project therefore requires biology, mathematics, and computer science to answer the same research question.

35 Final Summary

The combined project asks whether endometriosis changes the structure of RNA splicing in genes that may already be important because of genetic susceptibility or environmental response.

Instead of only measuring whether a gene is expressed more or less, we will represent its splice-junction structure as a weighted directed graph.

We will then use Hodge-theoretic methods to separate directional and cyclic components of the graph.

The overall question is:

Can mathematically measurable changes in RNA-splicing topology reveal differences between endometriosis and normal endometrial tissue, particularly in genes connected to genetic susceptibility and environmental response?

The final product will be a Rust program capable of performing this analysis on biological splice-junction data.

36 Short Version for Assignment Submission

Problem:

Endometriosis is influenced by complex biological factors that may include inherited genetic susceptibility, environmental response, and changes in gene regulation. Existing analyses often focus on identifying associated genes or measuring changes in overall gene expression. However, two samples can have similar expression of a gene while producing different RNA transcripts through alternative splicing.

Proposed Project:

We want to investigate whether genes associated with endometriosis show measurable differences in RNA-splicing structure between endometriosis and normal endometrial tissue. We will first prioritize genes associated with genetic susceptibility, environmental response, or both. We will then represent RNA splice junctions for these genes as weighted directed graphs and use graph-level Hodge decomposition to separate directional and cyclic components of the splice-junction signal. A Rust program will construct the graphs, perform the mathematical decomposition, compare disease and control samples, and identify genes with the largest changes in splicing topology. The goal is to determine whether endometriosis may involve changes not only in how much genes are expressed, but also in how their RNA transcripts are structurally assembled.